Do 11,000 Sharks Die Every Hour?

MA26M017

How unreported catches and finning drive staggering global mortality.



METHODS

Converted reported and estimated catch tonnages using specific and overall median weights.

KEY RESULT

Estimated 100 million global shark deaths in the year 2000.

TAKEAWAY

Fishing exploitation severely exceeds natural population rebound limits.

Media: BBC More or Less (Sep 2025) | **Data:** Worm et al. (2013)

Timestamps: 01:10 (Key Result) • 01:40 (Variety of Sharks) • 02:26 (Extinction Risk) • 02:55 (Methods) • 03:50 (Estimation) • 05:34 (Limitations)

Why are so many ex-prisoners being sent back to prison?

In England and Wales, prison recalls increased to 51,419 in the year ending March 2026, a 28% rise compared with the previous year. Many prisoners are recalled not for committing new crimes but for breaking release conditions. We aim to investigate the reasons behind this increase using hypothesis testing and logistic regression.

Objective: To investigate whether reoffending is associated with age group

- To test this hypothesis, the chi-square test of independence is carried out
- P-value=0.37
- Conclusion: Since P-value is greater than usual level of significance

There is insufficient statistical evidence of an association between age group and reoffending

Objective: To investigate whether reoffending is associated with housing condition.

- To test this hypothesis, the chi-square test of independence is carried out
- P-Value= 8.4×10^{-17}
- Conclusion: Since the P-value is smaller than usual level of significance

There is sufficient statistical evidence of an association between housing condition and reoffending

Conclusion: Housing condition, sentence length, and previous offences were significantly associated with reoffending. Homeless prisoners had approximately 5 times higher odds of reoffending, while each additional previous offence increased the odds by about 2.2%. Sentence lengths of 12 months or more were associated with lower odds of reoffending. However, age and gender were not statistically significant predictors in the model.

Objective: To study how different factors affect the probability of a prisoner reoffending after release.
For this purpose, we used a logistic regression model to examine how different factors are related to the probability of reoffending after release
The logistic regression model is as follows,

$$\ln\left(\frac{\hat{p}_i}{1-\hat{P}_i}\right) = 0.3951 - 0.0197G_i - 0.0028A_{1i} - 0.1862A_{2i} - 0.0922A_{3i} - 0.1456A_{4i} - 1.6313S_i + 1.6122H_i$$

Explanatory variable	Coefficient	Odds ratio	Interpretation
Female vs Male	-0.0197	0.981	1.9% lower odds
Age 21–30 vs 18–20	-0.0028	0.997	0.3% lower odds
Age 31–40 vs 18–20	-0.1862	0.83	17.0% lower odds
Age 41–50 vs 18–20	-0.0922	0.912	8.8% lower odds
Age 50+ vs 18–20	-0.1456	0.865	13.5% lower odds
Sentence 12+ months vs less than 12 months	-1.6313	0.196	80.4% lower odds
Previous offences (per additional offence)	0.02195	1.022	2.2% higher odds
Homeless vs Stable housing	1.6122	5.014	401.4% higher odds

CAN 8,000 TRACKED WEALTHY PEOPLE PREDICT 9,500 DEPARTING UK MILLIONAIRES?

Presented by: B Charulekha | NA23B041

News headline Rich ready to leave UK over budget tax threat

602,000 UK Millionaires → 9500 estimated departures



Observed dataset (Sample)

~ 50 net departures in Jan-May (5 months)

50 in 8,000 = 0.625%

Extrapolating to UK Millionaires population size (Jan-May)

UK Millionairs population = 602,000

0.625% of 602,000 = 3762.5

Extrapolating for the whole year of 2024

5 Months = 3762.5

for 12 Months, 3762.5 * 12 / 5 = 9,030

METHODS Track LinkedIn location changes in 8,000 people sample, extrapolate to 602,000 UK Millionares

KEY RESULT About 50 net leavers is projected to 9,500 annual departures from UK

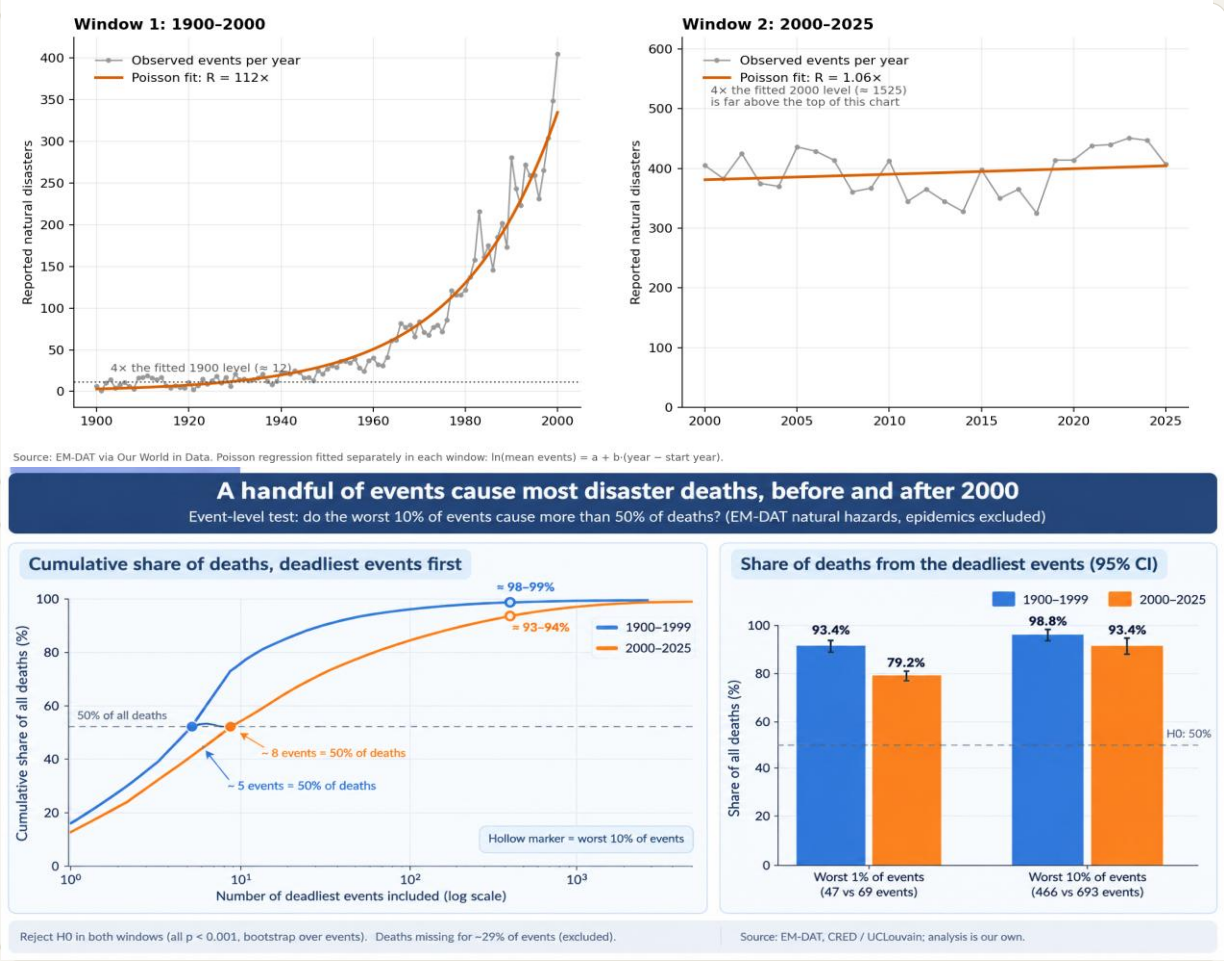
DATASET ISSUES Sample is skewed towards the ultra-rich population | Labour law changes causation is unsupported

How do you count millionaires? | More or Less | BBC | 18 Sept 2024

Relevant timestamps : Database [02:44] | UK millionaire population [02:17] | Net departure [03:18] | Data collection method [02:50] | Dataset issues [03:38]

Did Natural Disasters Really Quadruple?

We tested the claim on EM-DAT data.



Hypothesis 1 : Is the Rise in Recorded Disasters Truly Fourfold or more?

H_0: R ≤ 4 H1: R>4 ; Test: Wald z test on the slope of quasi poisson regression; α=0.05

1900–2000

Reject H_0

R = 112x

95% CI : (86 , 146)x

z = 24.6 , p < 0.001

2000–2025

Fail to reject H_0

R = 1.06×

95% CI: (0.94, 1.20)x

z = −20.9 , p ≈ 1

NOTE: EM-DAT(established in 1988) warns pre-2000 data is subject to reporting biases.

Hypothesis 2 : Do the deadliest 10% of disasters account for more than 50% of total deaths?

H_0: S<=0.5 H1: S>0.5 ; Test: Bootstrap hypothesis test **; α=0.05**

1900–1999

Reject H_0

98.8% - recorded deaths

95% CI: (97.6-99.3)%

2000–2025

Reject H_0

93.4% - recorded deaths

95% CI:(89.5,95.5)%

1900-1999: China's flood in 1931 caused 3.7M deaths

2000-2025: Haiti Earthquake in 2010 caused 223k deaths

Key result: Since 2000, we find no evidence of a fourfold increase in recorded disasters
-- A few extreme events drive disaster deaths.

Takeaway- The fourfold claim depends on the start year.

Source: BBC Radio 4, More or Less: Behind the Stats, [Title: "Are natural disasters on rise?"]. Data: EM-DAT (CRED / UCLouvain) via Our World in Data; tests are our own.

DO DIABETES-RELATED HEALTH INDICATORS SHOW THE SAME PATTERNS ACROSS TWO CATEGORIES OF INDIVIDUALS?

Comparing Age, Glucose and BMI using logistic regression and interaction testing

METHODS & MODELS

Two datasets: Pima Indians Diabetes and Indians Diabetes
Outcome (Y): diabetes status (0 = no, 1 = yes)
D: dataset indicator (0 = Pima, 1 = Prediction)

Reduced model (no interactions)

$$\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 D + \beta_2 A + \beta_3 G + \beta_4 B$$

Full model (with interactions)

$$\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 D + \beta_2 A + \beta_3 G + \beta_4 B + \beta_5 (D \times A) + \beta_6 (D \times G) + \beta_7 (D \times B)$$

A = standardized Age
G = standardized Glucose
B = standardized BMI
p = probability of diabetes

HYPOTHESES

Null hypothesis (H_0)

The Age, Glucose and BMI associations with diabetes are the same across both categories.

$$\beta_5 = \beta_6 = \beta_7 = 0$$

Alternative hypothesis (H_1)

At least one of these associations differs between the two categories.

$$\text{At least one of } \beta_5, \beta_6, \beta_7 \neq 0$$

Test: Likelihood Ratio Test comparing the reduced and full models (χ^2, 3 df).

RESULTS

Overall likelihood ratio test

p = 0.0236

Reject H_0 at α = 0.05

Individual interaction tests

Dataset × Age	**0.0022**	**Significant**
Dataset × Glucose	**0.4967**	**Not significant**
Dataset × BMI	**0.9317**	**Not significant**

Takeaway: Age shows the clearest individual difference. Glucose and BMI interactions are not individually significant.

Interpretation: evidence of differing associations between these datasets; this does not establish causation.

Inspired by: “Tale of Two Indians: Heterogeneity in Type 2 Diabetes Pathophysiology” (2019).

Are Refugees more likely to commit Crime ?

"Asylum seekers make up 0.8% of Dorset's population and 44% of alleged sex offenses "

Data Source	Offender	Claimed Rate	Actual
Social Media Post	Asylum Seekers	**44%**	Fabricated
Dorset Police FOI	Migrant Hotel Residents	N/A	**~1.0%(8 of 808 reported)[1]**
National Level FOI	Foreign Nationals	N/A	**7.0% - 8.0%**

Grok AI

" Dorset Police's FOI disclosure confirms that 51 of 116 males awaiting prosecution for sexual offenses in the Bournemouth area are asylum seekers from migrant hotels, which calculates to about 44% " [2]

KEY RESULT

1) **Viral 44% claim fabricated**; verified police data indicates 1%.

2) **AI Hallucination :** AI cited fake document to claim 44%. Admitted non-existence of data upon further prompting.

3) Most refugees from Afghanistan, Syria, Iran, and Eritrea come to the UK. They usually arrive through government programs/asylum system. **War is the main reason.**

4) Demographic skew as most asylum seekers are young

Timestamps
(1) - 1:41 (Investigated for crimes), (2) - 3:48 (Conversation with Grok), (3) -5:16 , 6:04 (Conversation with Grok)

Alcohol and heart health: hypotheses and results

A: Blood Pressure

H_0 : $\mu_{\text{drinkers}} = \mu_{\text{non-drinkers}}$

H_1 : $\mu_{\text{drinkers}} > \mu_{\text{non-drinkers}}$

Systolic BP. One-sided Welch t-test (unequal variance)

B: Heart Disease

H_0 : $p_{\text{drinkers}} = p_{\text{non-drinkers}}$

H_1 : $p_{\text{drinkers}} \neq p_{\text{non-drinkers}}$

Heart disease. Two-proportion z-test ($n = 68{,}610$)

C: Only Non-smokers

H_0 : $p_{\text{drinkers}} = p_{\text{non-drinkers}}$

H_1 : $p_{\text{drinkers}} \neq p_{\text{non-drinkers}}$

Same z test as B on non-smokers ($n = 62{,}574$),

	Drinkers	Non-drinkers	Result ($\alpha = 0.05$)
A: mean systolic BP(mmHg)	129.2	126.5	$\lvert T_s\rvert = \lvert 8.68\rvert > 1.96,\ p < 0.001 < 0.05$
B: share with heart disease	47.7%	49.6%	$\lvert Z_s\rvert = \lvert -2.245\rvert > 1.96,\ p = 0.024 < 0.025$
C: heart disease in non-smokers	51.3%	49.7%	$\lvert Z_s\rvert = \lvert 1.421\rvert < 1.96,\ p = 0.155 < 0.025$

Conclusion.

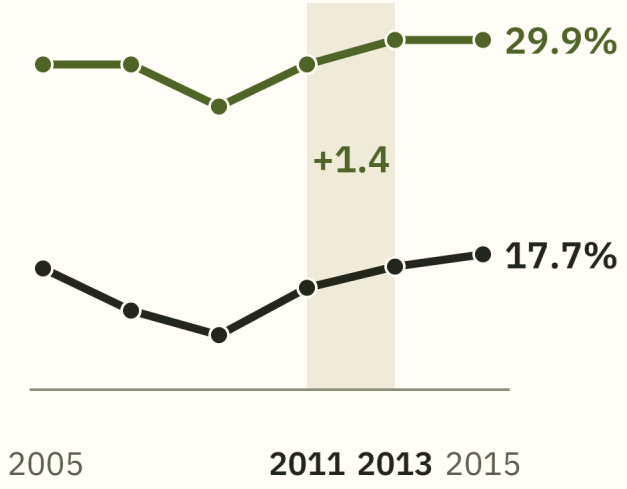
- Drinkers have higher systolic BP (reject H_0). It holds among non-smokers too.
- Drinkers show slightly less heart disease (reject H_0). This does not mean alcohol protects the heart.
- Among non-smokers, drinkers have slightly more heart disease(fail to reject H_0).
- Alcohol drinking is significantly associated with higher blood pressure, but no reliable evidence for changes the heart disease rate. Drinking is a yes/no with no amounts so there are association, not proof of cause.

Cardiovascular Disease dataset (OpenML 45547), about 68,600 records after cleaning.

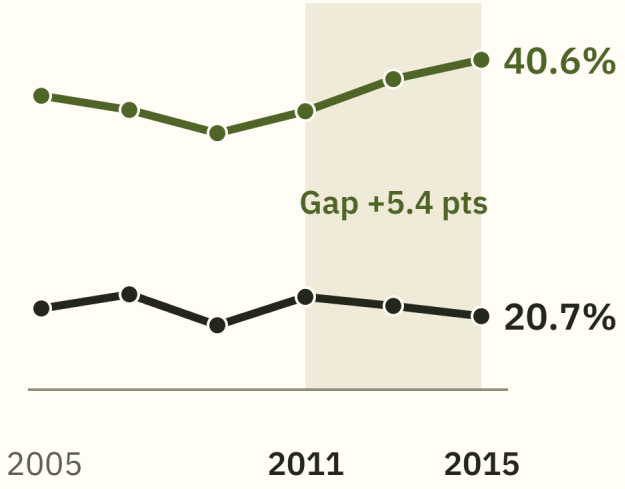
Did social media cause the teen mental health crisis?

Teen hopelessness rose gradually, mostly among girls; is social media the cause?

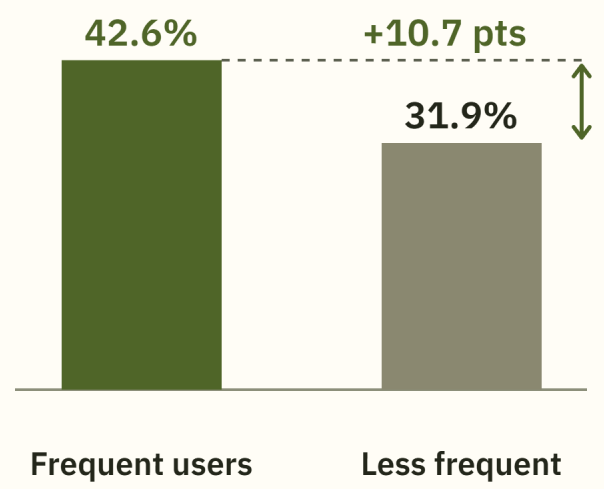
1 · Cliff or slow climb?



2 · Girls vs boys, felt hopeless



3 · Heavy vs lighter use



METHODS

Z-tests and difference-in-differences on CDC survey data; prevalence ratio for 2023.

KEY RESULT

Gradual rise; girls +3.95 vs boys −1.45; heavy users 42.6% vs 31.9%.

TAKEAWAY

Gradual, girl-driven rise; social media linked, but cause unproven.

"Jonathan Haidt Explains the Youth Mental Health Crisis," The Screenagers Podcast, Jun 10, 2024 · 03:52 · 05:57 · 09:04

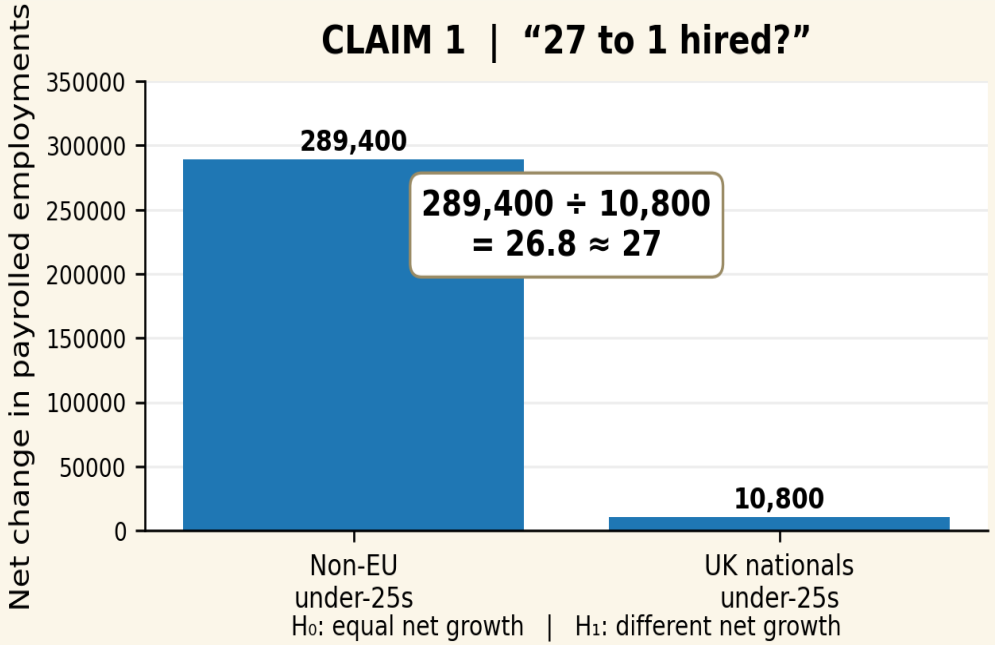
Data: CDC Youth Risk Behavior Survey (cdc.gov/yrbs)

WHEN HEADLINES MEET DATA: TWO STATISTICAL CHECKS

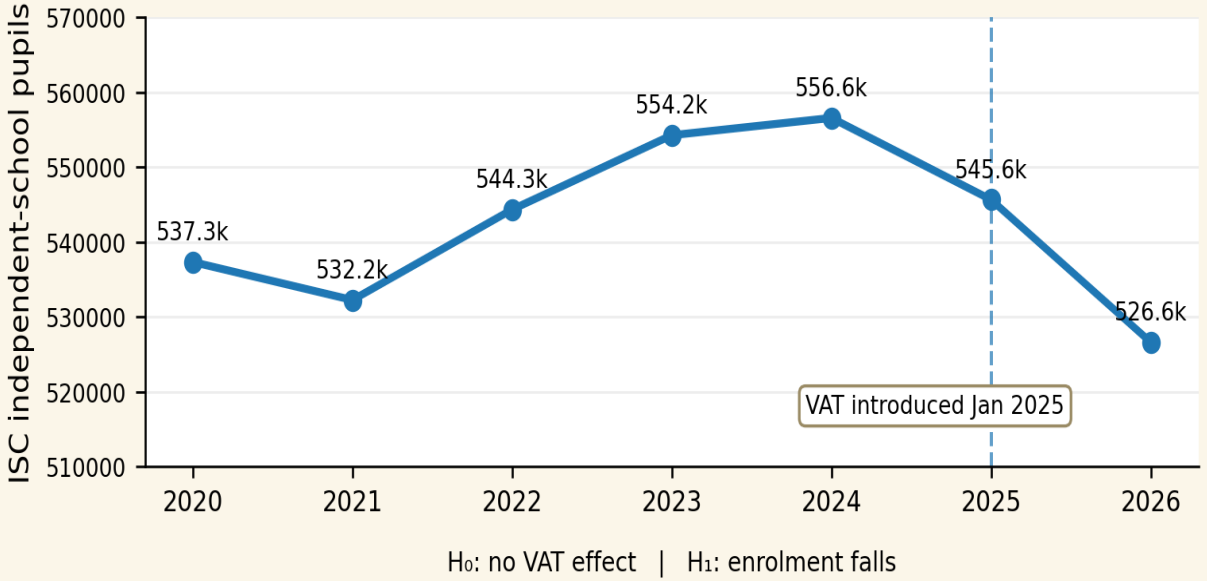
A number can be correct—and still tell the wrong story.

BBC Radio 4 | More or Less: "Education, Education, Education (and immigration)" | 10 June 2026

CLAIM 1 | "27 to 1 hired?"



CLAIM 2 | "Did VAT reduce enrolment?"



METHODS

Compare payroll growth; test VAT impact using Difference-in-Differences.

KEY RESULT

Ratio equals 26.8; pupil totals fell 1.96% after 2024.

TAKEAWAY

Net job growth is not hires; trends alone prove no causation.

LIMIT: descriptive figures only; matched population data and state-school comparisons are needed for formal tests.

SOURCE: BBC Radio 4, More or Less, "Education, Education, Education (and immigration)," 10 Jun 2026; 00:32 (employment), 08:40 (VAT). Data: HMRC payroll figures (CSJ analysis); ISC Census 2020–2026. 95% CI/DiD are proposed, not completed tests.

STATISTICS & INFERENCE | BBC RADIO 4: MORE OR LESS

THE STATS OF THE NATION

How do we evaluate claims about immigration, benefits and inequality?

DESCRIPTIVE STATISTICS → summarise observed data | STATISTICAL INFERENCE → assess what the data supports

1. IMMIGRATION

98% claim

Descriptive: calculate the share of population growth associated with net migration.

Inference: check the population estimates, time period and uncertainty before generalising.

2. BENEFIT SPENDING

0.1% of GDP

Descriptive: express unemployment-benefit spending as a proportion of GDP and compare countries.

Inference: ask whether countries use comparable definitions and datasets.

3. INCOME INEQUALITY

14.5% top 1% share

Descriptive: measure the income share of the richest 1%; compare it over time.

Inference: consider data coverage, capital gains and whether the trend is robust.

A SIMPLE INFERENCE CHECK

Claim → define the measure → identify the data/sample → quantify uncertainty → draw a cautious conclusion

Source: BBC Radio 4, More or Less, “The Stats of the Nation” (9 Jan 2026). Figures are episode-reported and depend on definitions/time periods.

MA 5104 | STATISTICAL INFERENCE & LEARNING

ARE INDIANS REALLY HAPPY?

India’s rank is low—but what do the numbers really reveal?

1 | WHAT WAS MEASURED?

4.389 / 10

India’s mean life evaluation

RANK 118 of 147 countries

0 = worst life | 10 = best life

2022–2024 average • Gallup World Poll

95% rank interval: 114–129

2 | INDIA: INDICATORS REPORTED IN ARTICLE

23rd	Personal freedom
93rd	GDP per capita
56th	Perceived corruption
60.6 y	Healthy life expectancy
58%	Have someone to rely on

Article-reported context; not all determine the rank.

3 | HOW THE ANALYSIS WORKS

GALLUP SURVEY → CANTRIL LADDER

Gallup asks people to rate their life from 0–10.

3-YEAR AVERAGE (2022–24)

Uses a pooled 3-year average (2022–24).

CROSS-COUNTRY OLS REGRESSION

Models six factors: income, support, health, freedom, generosity and corruption.

4 | REGRESSION EVIDENCE (WHR 2025, TABLE 2.1)

Outcome: Cantril Ladder score | Pooled OLS, 155 countries, 2,234 observations

Social support	+2.686	**Freedom**	+1.518	**GDP**	+0.328

Associations, not proof of cause and effect.

5 | INTERPRETATION & LIMITATION

A national average hides differences across people and regions.

A survey score cannot capture every lived experience.

India's happiness ranking is not meaningless, but neither is it the complete story of Indian happiness.

Sources: World Happiness Report 2025, Fig. 2.1 & Table 2.1 (Gallup World Poll); Maya Pillai, “India World Happiness Report 2025: Are Indians Really Unhappy?”

IS THE PLANET GETTING GREENER?

BBC MORE OR LESS | PODCAST SUMMARY

4.6%	20%	23%
GLOBAL LEAF AREA	**CHINA'S LEAF AREA**	**ARID + SEMI-ARID**
Approx. increase over 20 years	Approx. increase over 20 years	Share of global greening

HOW IS GREENING MEASURED?

Satellites analyse light reflected and absorbed by plants. Scientists use the **Leaf Area Index (LAI)** to estimate leaf area relative to ground area.

WHY IS IT HAPPENING?

CO2 fertilisation • Tree planting • Longer growing seasons in some colder regions • Agricultural changes

WHAT ABOUT DRY REGIONS?

Most greening occurs in wetter regions. The 23% figure means arid and semi-arid regions account for a share—not the majority—of global greening.

WHY IS MORE CO2 NOT ALL GOOD?

Climate change raises risks from heatwaves, droughts and sea-level rise, harms ecosystems and can reduce crop yields. Plant growth cannot cancel human-caused emissions.

TAKEAWAY: Earth is greening, but the global increase is about 4.6%—not 20%.
A greener planet is not necessarily a healthier planet.



Are Women 14 Times More Likely to Die in Natural Disasters?

Let's look at the real data.



Data and Statistical Methods

Compare mortality across countries.

What the Research Says

Results vary by country & disaster.

Real Examples of Natural Disasters

Bangladesh Cyclone (1991)	Haiti Earthquake (2010)	Hurricane Katrina (2005)
= 4.7×	= 0.9×	= 0.9×

Limitations of the Research

- No strong uniform evidence.
- Results differ by region.

Real-Life Applications

Helps improve preparedness & policy.

Result:

Death rates vary — not **14×**.

Takeaway:

A claim cannot be established by just observation, but through **proper evidence**, but even in such cases we should account for **other factors**.

- Not just women — sometimes children too.
- Men are higher risk in developed countries; women in developing countries.

Source: BBC, "Are women 14 times more likely to die in natural disasters?", Oct. 30, 2018.

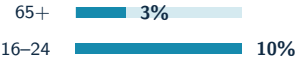
Is There Really a Loneliness Epidemic?

THE QUESTION Young people report loneliness more often than older people. Does that prove loneliness is increasing over time?

01 AGE SNAPSHOT

England · ONS Community Life Survey, 2016–17

Often / always feel lonely



Five response options: often/always, sometimes, occasionally, hardly ever, never.

02 CHANGE OVER TIME

US high-school students · 1977–2012

Mean score from six loneliness items

Scale: 1 = disagree, 5 = agree



Schematic trend only; the source chart reports a very small decline. Its vertical axis is truncated.

03 THREE WAYS TO ANALYSE

Cross-sectional

Compare age groups at one time. Describes differences, not trends.

Longitudinal

Follow the same people as they age.

Cohort comparison

Compare gen at similar ages.

Source:https://ourworldindata.org/loneliness-epidemic

Should You Follow the 5-Second Rule?

G 30

Does faster pickup prevent bacterial contamination?

Methods & Analysis

Statistical Analysis (t-Test)

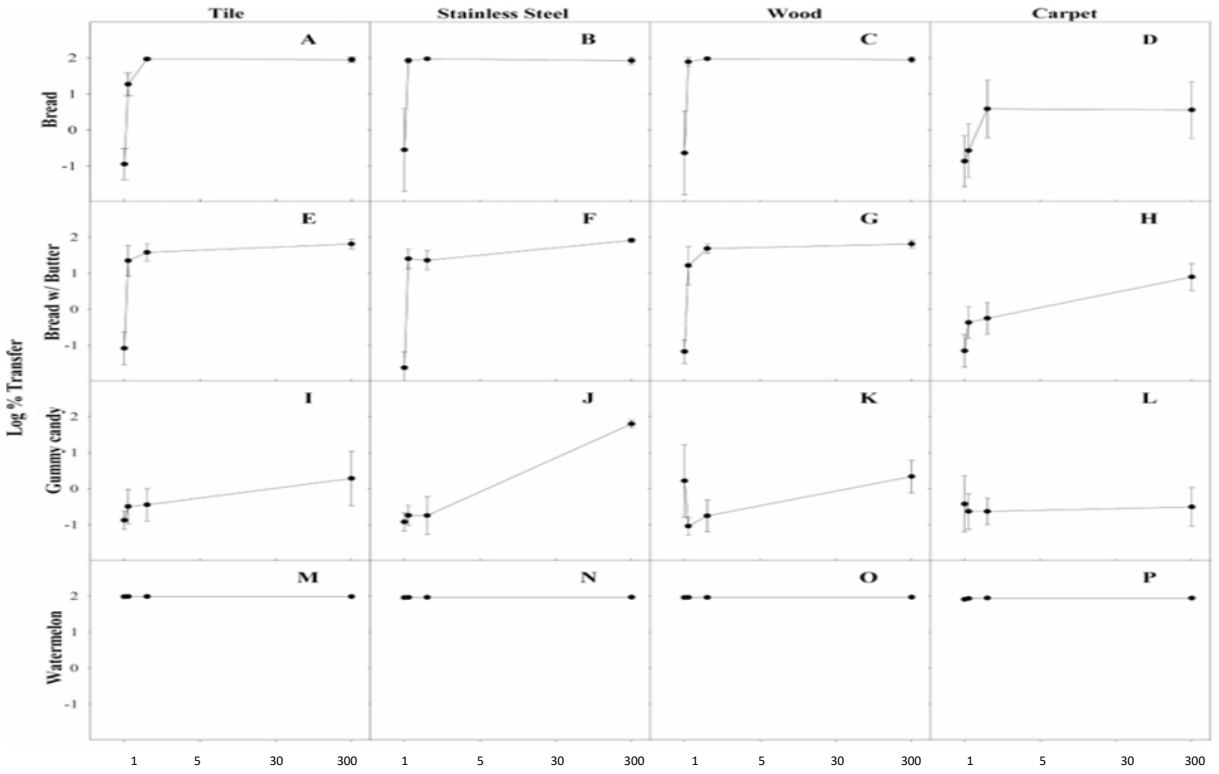
Computed sample mean ($\bar{X}$) and sample variance (S^2) for each food group. Used two-sample t-tests to evaluate whether differences across contact times are statistically significant.

Method & Sampling

Tested 4 food types (watermelon, bread, bread with butter and gummy candy) dropped on contaminated surfaces across 4 contact durations(0s control, 5s, 30s, 300s) with measured bacterial counts (CFU/g).

Primary Variables / factors

Contamination levels depend heavily on four factors: contact time, food type, surface type, and the bacterial preparation method.



Limitations

Conducted in controlled laboratory conditions with specific food items and bacteria. Lab counts do not directly equate to infectious dose or actual illness risk in home environments.

Key Results

Bacterial contamination increases with contact time (Watermelon> bread with butter> bread > Gummy candy). Bacterial transfer occurs instantaneously (<1s),disproving the absolute 5-second safety rule.

Takeaway

The 5-second rule is not a simple yes/no threshold. Picking up food faster reduces relative contamination, but surface cleanliness and food moisture dictate real safety.

Source: BBC Radio 4, More or Less, "Should You Follow the 5-Second Rule?" (2022).
Audio timestamps: Intro 0:30-2:45 | Key finding / Data 22:50-28:15

Miranda, R. C., & Schaffner, D. W. (2016). Longer contact times increase cross-contamination of Enterobacter aerogenes from surfaces to food. https://doi.org/10.1128/AEM.01838-16

The Medical Test Paradox & Bayesian Inference

UDIT SRIVASTAVA
ADITYA MAHANTA

1. DEFINITIONS & BASELINE COHORT

Prevalence / Prior P(D): Base rate of disease in population = **1%** (10 in 1,000).

Sensitivity P(+|D): True positive rate; test accuracy given disease = **90%**.

Specificity P(-|~D): True negative rate; accuracy given healthy = **91%**.

False Positive Rate P(+|~D): Healthy testing positive = **9%** (100% - Specificity).

PPV P(D|+): Positive Predictive Value; actual chance of disease given positive result.

1,000-Woman Cohort Frequency Breakdown:

Test Result	Cancer (10)	Healthy (990)	Total
Positive (+)	**9 (True +)**	**89 (False +)**	**98**
Negative (-)	1 (False -)	901 (True -)	902

Core Problem: Because disease is rare (1%), the total false positives (89) far outweigh true positives (9).

2. MATHEMATICAL CALCULATIONS

METHOD A: STANDARD BAYES' RULE (PROBABILITIES)

$$P(D|+) = [\ P(+|D) \times P(D)\ ] / P(+)$$

- Numerator = 0.90 × 0.01 = **0.009**
- Denominator = (0.90×0.01) + (0.09×0.99) = **0.0981**
- P(D|+) = 0.009 / 0.0981 = **9.17%**

(Frequency table gives 9 / 98 = **9.18%** due to rounding 89.1)

METHOD B: REDESIGNED ODDS FORM

$$\text{Posterior Odds} = \text{Prior Odds} \times \text{Bayes Factor}$$

- **Prior Odds:** Odds of disease before test - [**P(D)/P(-D)**].
- **Bayes Factor:** How much a positive result shifts the odds (**sensitivity/ false-positive rate**).
- **Posterior Odds:** Odds of disease after a positive result.

- **Prior Odds:** 1% prob = **1 : 99**
- **Bayes Factor:** Sens / FPR = 90% / 9% = **10**
- **Posterior Odds:** (1 : 99) × 10 = **10 : 99**
- **Posterior Prob:** 10 / (10 + 99) = **9.17%**

Gigerenzer Seminar Finding: >50% of practicing gynecologists incorrectly answered 90%, confusing Sensitivity P(+|D) with PPV P(D|+).

3. CONTEXT SHIFT & APPLICATIONS

Impact of Base-Rate Prevalence Shift:

- **At 1% Prevalence:** Prior 1:99 × 10 = 10:99 Posterior = **9.17% PPV**
- **At 10% Prevalence:** Prior 1:9 × 10 = 10:9 Posterior = **52.6% PPV**
- ***Insight:*** *Diagnostic value changes radically based on patient population.*

Critical Evaluation & Limitations:

- **Hypothetical Model:** Demonstrates mathematical principles rather than direct real-world patient outcomes.
- **Dynamic Parameters:** Prior prevalence and sensitivity vary by age, symptoms, and clinical setting.

Cross-Domain Applications:

- **Fraud Detection:** Rare fraud leads to high false alarms despite high classifier accuracy.
- **Airport Security:** Rare threats mean alerts require contextual verification.

Takeaway: Always report PPV/NPV alongside target population prevalence.

Key Takeaway: A positive test is evidence that updates prior belief, not a final diagnosis.
To interpret evidence correctly, combine test accuracy (Bayes factor) with prior risk (base rate).

Posterior Odds = Prior Odds × Bayes Factor

Does the news reflect what we die from?

Actual causes of death vs. media coverage in the US (2023)

ACTUAL DEATHS

56%

Heart disease (29%) + Cancer (27%)

NYT MEDIA COVERAGE

7%

Heart disease (29%) + Cancer (27%)

Where does media attention go?

Share of media coverage devoted to each cause

News outlet	HOMICIDE	TERRORISM
New York Times	42%	18%
Washington Post	46%	12%
Fox News	52%	11%

Overrepresentation in NYT coverage

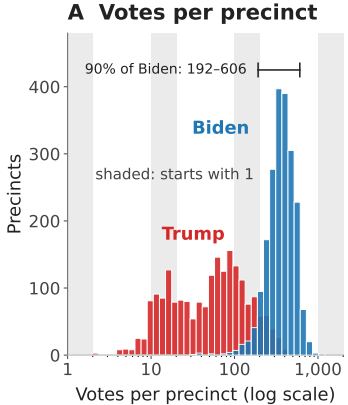
43× HOMICIDE

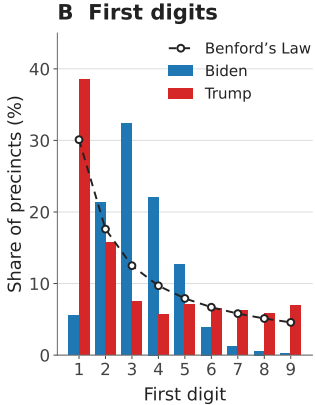
18,240× TERRORISM

15 selected causes account for approximately 76% of US deaths in 2023.
Source: US CDC, Media Cloud, Global Terrorism Index; Ritchie et al. (2025), Our World in Data.

Do Chicago's 2020 votes really break Benford's Law?

In 2020, people claimed Biden's Chicago votes were fake because they don't follow Benford's Law.





Methods

First-digit and last-digit frequency checks on Chicago precinct vote counts.

Key result

Precincts are similar in size, so Biden's first digits can't follow Benford.

Takeaway

Benford's Law can flag odd data, but it can't prove fraud.

Media: M. Parker, *Why do Biden's votes not follow Benford's Law?*, Stand-up Maths, YouTube, 10 Nov 2020 (youtu.be/etx0k1nLn78)
Timestamps: 0:01–3:40 Benford's Law · 3:41–6:26 forensic accounting · 6:27–10:49 Chicago election data · 10:50–15:46 last-digit check
Figure: re-plotted in Python from Chicago Board of Election Commissioners 2020 precinct results (2,069 precincts)

EXPECTED GOALS & STATS IN FOOTBALL | THE TRENDS

A scoreline tells us what happened - xG helps explain the quality of chances created.

What xG means?

Expected Goal (xG): Probability that a shot will become a goal.
Range: 0 (Lower) ----------1 (Higher)
xG of 0.7 = 7 goals out of 10 similar shots

How is xG calculated?

- Shot distance & angle
- Velocity & Trajectory
- Shot type
- Shot position wrt the ground
- Defenders, attackers & keeper position

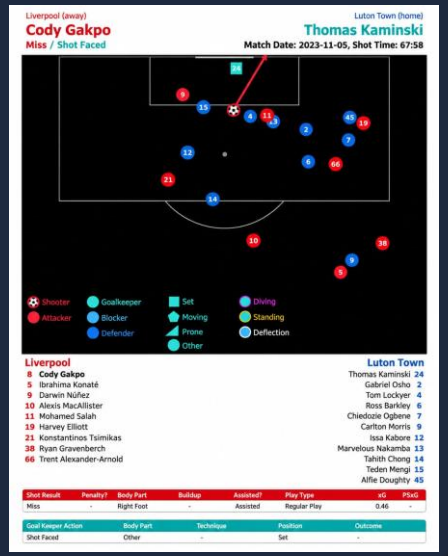


Limitations

- xG measures chances, not certainty
- Model definitions differ
- Some actions are unmeasured.

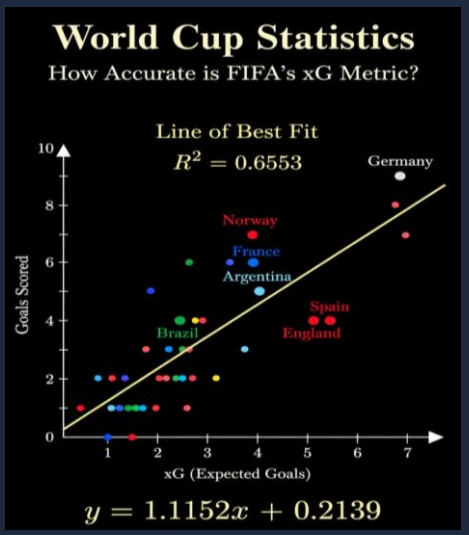
Same Area ≠ Same Chance



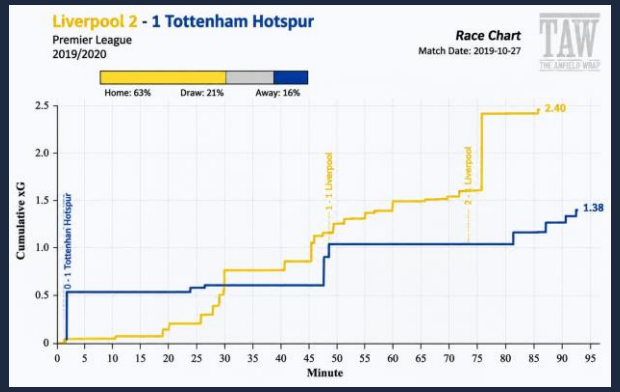


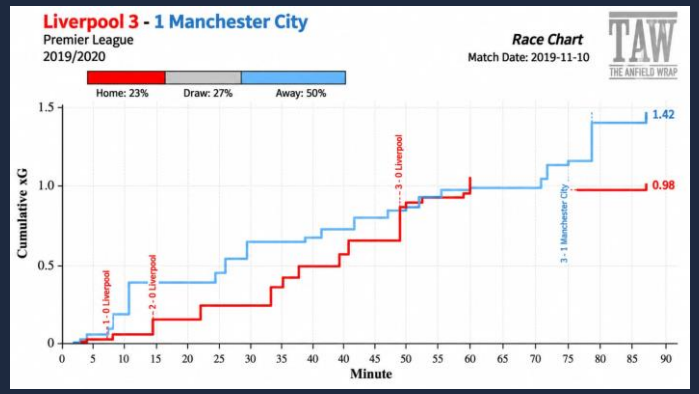
Source: Statsbomb

The Trends



Race Chart





Source: The Anfield Wrap

xG → Scorelines → Match Context → Longer-term Performance Trends

From One Shot To A Season-long Trend

- Over many matches, xG helps reveal performance trends that the scoreline can hide.
- Predicting future team performance.
- Across matches, xG trends can reveal more than one scoreline.

Key Takeaway

- Use xG with match context and observation - not as a verdict.

Media: The Anfield Wrap (TAW) "Expected Goals & Stats In Football, The Trends" | YouTube & Spotify | 24 July 2024 | ~24min

By: Eric P. Karr [CE25M827] & Gemechis Teferi Benti [CE25M812]

Can Random Selection Explain the Absence of Conservative MPs at PMQs?

Could none of 47 Conservatives appear among 15 names by chance?

SHUFFLE DATA | **311** applicants | **47** Conservative | **264** Non Conservative | **15** randomly selected

TWO METHODS, ONE PROBABILITY

Hypergeometric distribution

Count favourable selections / all possible selections

$$P(X = 0) = C(264, 15) / C(311, 15)$$

Sequential conditional probability

Multiply the changing probabilities without replacement

$$P(X = 0) = (264/311) \times (263/310) \times \ldots \times (250/297)$$

SAME RESULT **8.05%** ≈ 1 in 12 comparable draws

METHODS

Hypergeometric counting and conditional probabilities yield identical results.

KEY RESULT

Zero Conservatives among 15 selections has an 8.05% chance.

TAKEAWAY

Same model audits admissions lotteries, conditional on who applies.

Source: More or Less, “Does Europe suffer more heat deaths than the US?”, BBC Radio 4, 17 Jun 2026; PMQs segment 14:46–19:01, calculation 16:22–18:28. Shuffle procedure: UK Parliament.

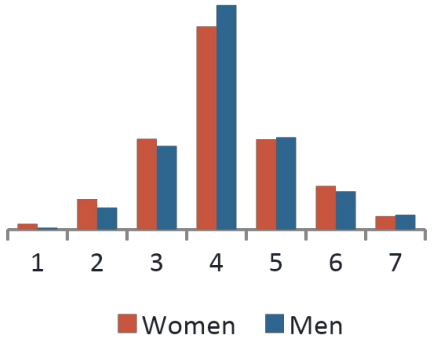
Is “Women’s Winter” Real? Testing a Podcast Claim with Data

Ayush Sourav Jagaty (MA26M007) • Roki Howladar (MA26M016) • MA 5104 Statistical Inference and Learning, IIT Madras

Do women really vote colder in offices, or do they just sit in colder rooms?

Welch t-test: vote by gender

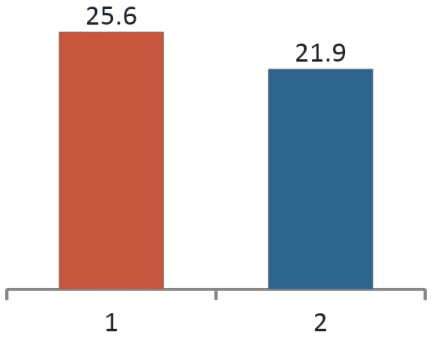
H_0: mean vote, women = men
H_1: mean vote, women ≠ men



Mean vote 0.101 vs 0.165 (t = −5.62, p < 0.001): reject H_0,

Chi-squared: feels cold

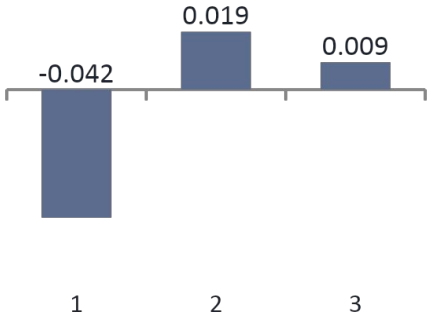
H_0: cold is independent of gender
H_1: cold depends on gender



$\chi^2 = 72.4$, $p < 0.001$: reject H_0.

Regression: female effect

H_0: β(female) = 0, adjusted
H_1: β(female) ≠ 0, adjusted



Women’s rooms ran 0.55°C cooler. Adjusted, p = 0.42: cannot reject H_0

Methods

Welch t-test, chi-squared test, and regression on ASHRAE office data (n=38,403).

Key result

Women feel colder (25.6% vs 21.9%); adjusting for room temperature erases it.

Takeaway

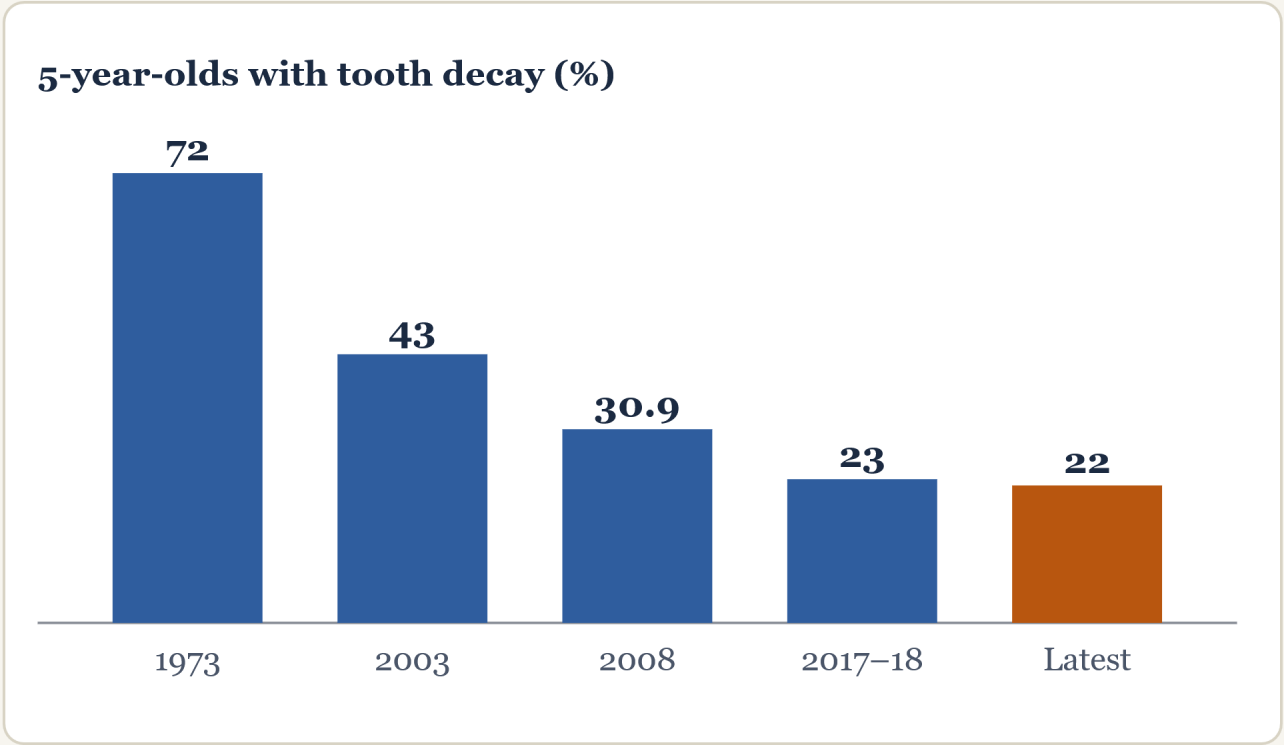
Real but small, mostly cooler rooms; not true for every woman.

Limits: *observational pooled surveys; metabolic rate assigned, not measured; podcast lab study had only 16 women .*
Fix: compare within the same building.

Media: BBC Radio 4, More or Less, “Saturday at 15:08” Data: ASHRAE Global Thermal Comfort Database II, office rows with recorded gender (n = 38,403).

Bad teeth and A&E: separating hospital admissions from A&E attendances

An MP's claim mixed up two different counts (02:00–02:40).



Methods: Compare admission rates per 100,000 with school dental survey percentages

Key result: Claim confused admissions with A&E; decay tops admissions but is falling

Limits: Little A&E data on dental causes; old surveys not directly comparable

Use elsewhere: Any "number one cause" claim: road injuries, school absence

Takeaway: Check what a headline counts before trusting the number

Source: More or Less, BBC, 16 Sep 2026 · Timestamps: 00:40–01:00 · 02:00–02:40 · 04:00–04:40 · 05:40–07:20

GDP and India's Economic Well-being

A Statistical Analysis

Jitendra Kumar (MA26M010) & Aditya Kumar (MA26M001)

1. Descriptive Statistics

$n = 26$ (Number of observations)

$\bar{g} = 6.25\%$ (Mean growth) — Med $= 7.31\%$ (Median)

$s = 2.98$ pp (Standard deviation)

Min: -5.78% (2020) — Max: 9.69% (2021)

2. Hypothesis Testing

Compare mean GDP growth across periods:

$$H_0 : \mu_1 = \mu_2 \qquad H_1 : \mu_1 \neq \mu_2$$

Statistic	Value
Mean (2000–2010)	6.49%
Mean (2011–2020)	5.21%
Welch's t	0.888
p-value	0.391

Decision: Fail to reject H_0 at $\alpha = 0.05$.

3. Linear Regression

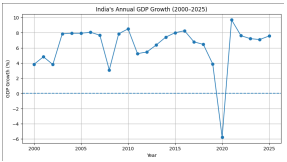
$$g_t = \beta_0 + \beta_1 t + \epsilon_t$$

Slope: 0.00345 percentage points/year — $R^2 = 0.000078$, $p = 0.966$

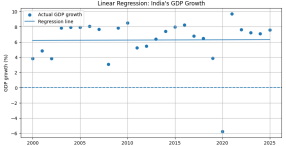
4. Preliminary Findings

- **Period Means:** No statistically significant difference in mean GDP growth across periods ($p = 0.391$).
- **Linear Trend:** No significant long-term upward or downward trend over time ($p = 0.966$).

5. GDP Growth Over Time



6. GDP and Well-being



Source: World Bank, World Development Indicators (WDI). Annual observations may be temporally dependent.

Why Cricket Defeats Hurt Indian Markets More Than Wins Help

India won the 2026 T20 World Cup final, yet markets fell nearly 2% on Monday.

Average next-day Nifty fall, by match result



Loss effect: over 7× the win effect (06:40–06:52)

SAME MONDAY, OTHER SHOCKS

Middle East escalation (01:42), Korea –10% (02:06), oil $115 (02:12), rupee past 92 (02:18)

METHODS

Next-day market returns compared after wins, losses and rivalries.

KEY RESULT

Losses move Indian markets over seven times more than wins.

TAKEAWAY

Sentiment moves markets, but confounders must be ruled out first.

Source: “Raghav’s Take”, The Quint, March 2026. Chart redrawn from on-screen Hindustan Times figures (win “flat”, loss 0.231%).
Timestamps: 02:22 · 04:02–04:38 · 05:00–06:06 · 06:40–06:52 · 07:08–08:08 · 09:02–10:16